

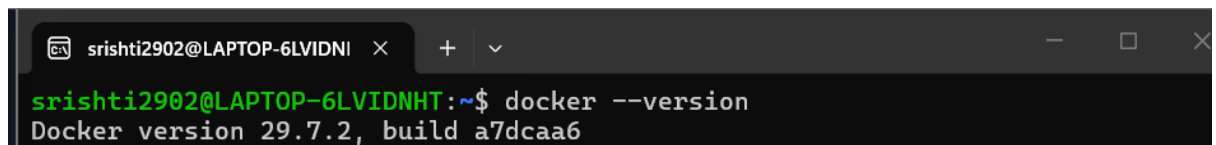
Lab 1: DevOps Foundations & Continuous Integration

1. Record all commands

Record all commands, pipeline configurations, screenshots, commit history, and observations for each experiment.

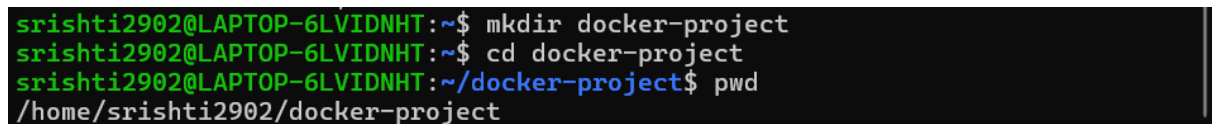
Docker Commands

1. Check Docker Installation: `docker --version`

A terminal window with a dark background. The title bar shows 'srishti2902@LAPTOP-6LVIDNI'. The prompt is 'srishti2902@LAPTOP-6LVIDNHT:~\$'. The command 'docker --version' has been entered, and the output is 'Docker version 29.7.2, build a7dcaa6'.

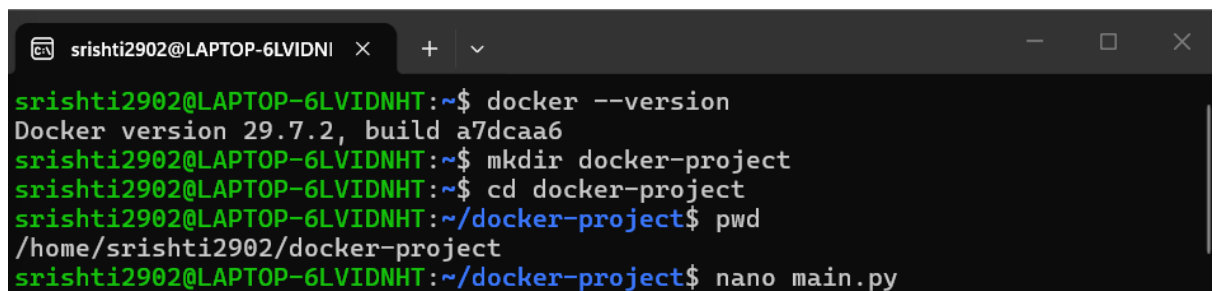
```
srishti2902@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
```

2. Create Project Directory: `mkdir docker-project`, `cd docker-project`, `pwd`

A terminal window showing the creation of a directory and navigation. The prompt is 'srishti2902@LAPTOP-6LVIDNHT:~\$'. The commands 'mkdir docker-project', 'cd docker-project', and 'pwd' are entered in sequence. The output of 'pwd' is '/home/srishti2902/docker-project'.

```
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir docker-project
srishti2902@LAPTOP-6LVIDNHT:~$ cd docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/srishti2902/docker-project
```

3. Create Python Application: `nano main.py`

A terminal window showing the execution of the nano command. The prompt is 'srishti2902@LAPTOP-6LVIDNHT:~/docker-project\$'. The command 'nano main.py' is entered. The previous commands and their outputs are visible in the history.

```
srishti2902@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir docker-project
srishti2902@LAPTOP-6LVIDNHT:~$ cd docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/srishti2902/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
```

```
srishti2902@LAPTOP-6LVIDNI x + v
GNU nano 8.7.1 main.py *
from flask import Flask

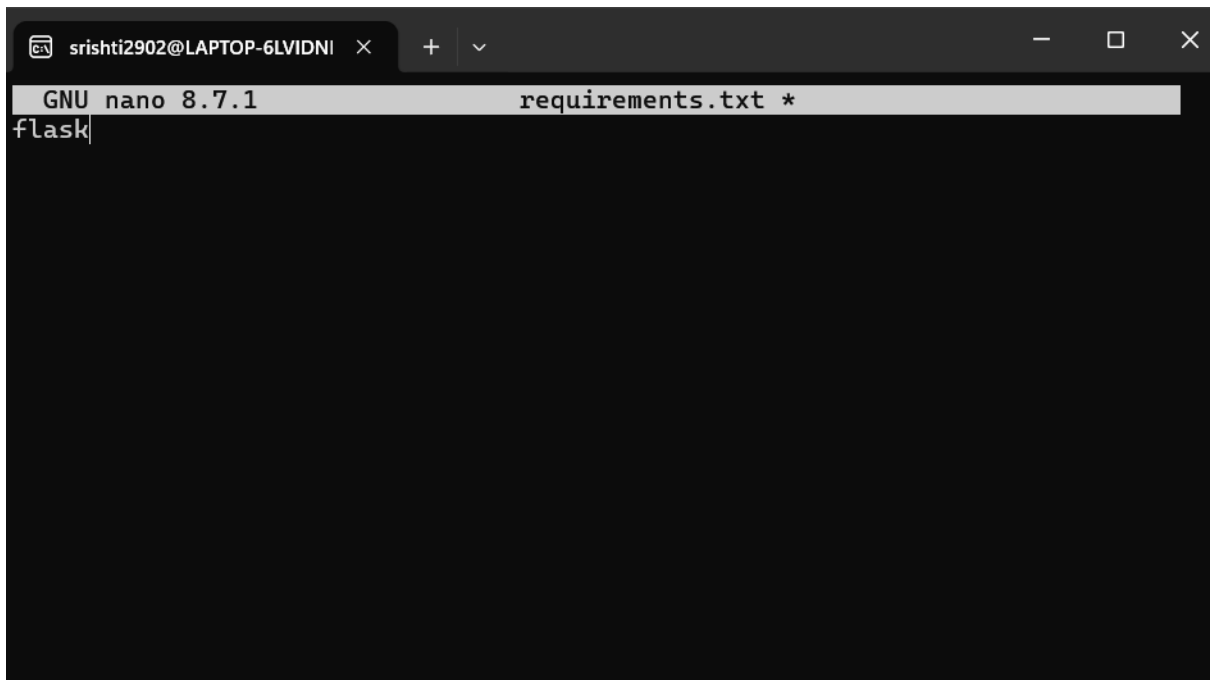
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello! My Docker application is running."

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)|
```

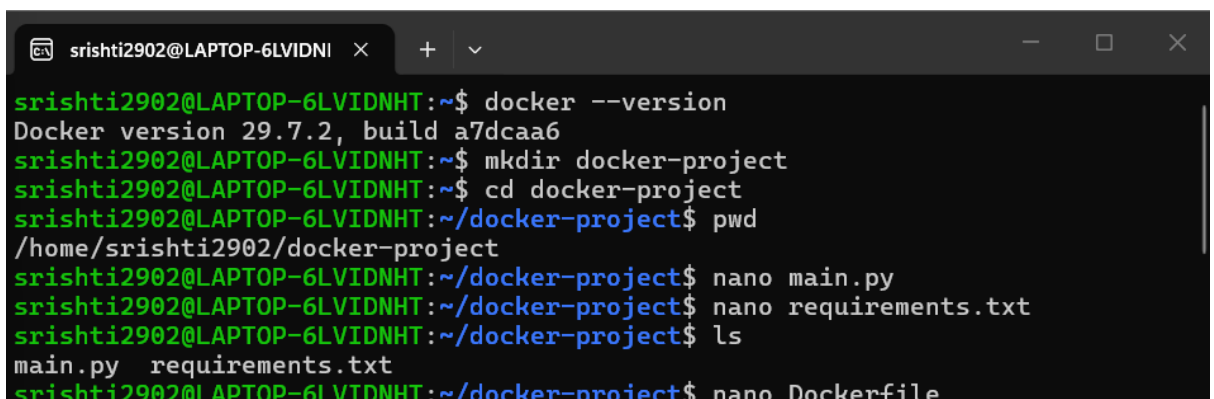
4. Create Requirements File: nano requirements.txt

```
srishti2902@LAPTOP-6LVIDNI x + v
srishti2902@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir docker-project
srishti2902@LAPTOP-6LVIDNHT:~$ cd docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/srishti2902/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano requirements.txt
```



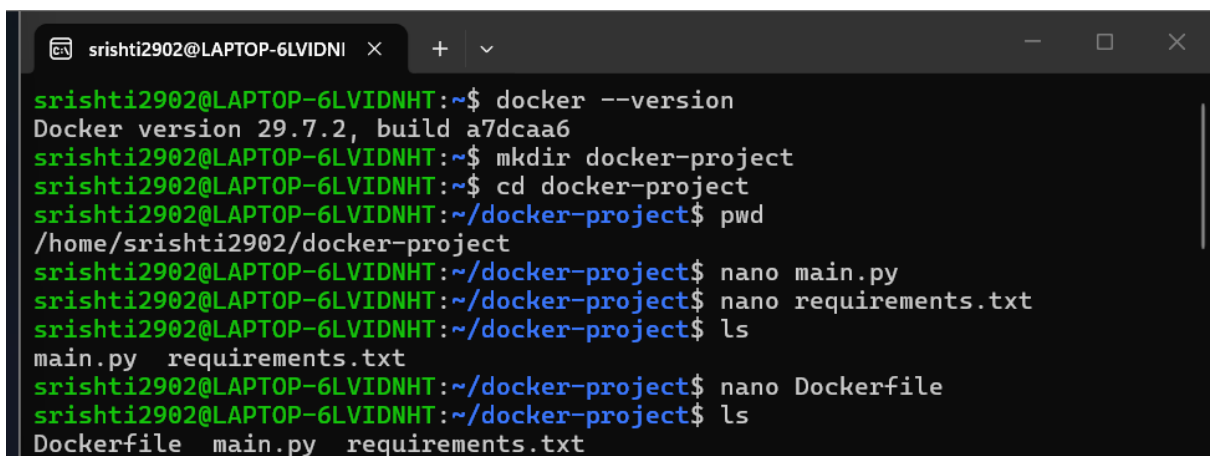
```
GNU nano 8.7.1 requirements.txt *
flask
```

5. **Create Dockerfile:** nano Dockerfile



```
srishti2902@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir docker-project
srishti2902@LAPTOP-6LVIDNHT:~$ cd docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/srishti2902/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
main.py requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano Dockerfile
```

6. **List Project Files:** ls



```
srishti2902@LAPTOP-6LVIDNHT:~$ docker --version
Docker version 29.7.2, build a7dcaa6
srishti2902@LAPTOP-6LVIDNHT:~$ mkdir docker-project
srishti2902@LAPTOP-6LVIDNHT:~$ cd docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ pwd
/home/srishti2902/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
main.py requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano Dockerfile
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile main.py requirements.txt
```

7. **Build Docker Image:** docker build -t my-docker-app .

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker build -t my-docker-app .
[+] Building 76.4s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 203B                               0.0s
=> [internal] load metadata for docker.io/library/python:3.13     7.0s
=> [auth] library/python:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/5] FROM docker.io/library/python:3.13@sha256:a05a0a9cfe9db7d6 55.7s
=> => resolve docker.io/library/python:3.13@sha256:a05a0a9cfe9db7d6a 0.1s
=> => sha256:1599fa592074badc651117d15e0d7cc070b21d3 6.09MB / 6.09MB 2.3s
=> => sha256:c5b14f510815b7cd08ec1fcfa5a5222e3520 27.66MB / 27.66MB 10.6s
=> => sha256:7af818fe6364cc23e3187871665225f2ed1ffd6f965 250B / 250B 0.9s
=> => sha256:44fed46b68cf907ddf6afb6bc4211842f0 236.34MB / 236.34MB 43.3s
=> => sha256:00f78834a2fef03250be89a7741ff39d0e6bb 25.64MB / 25.64MB 8.3s
=> => sha256:cbc19164244e861d91ebd80a17e2c78c5be4 67.80MB / 67.80MB 21.6s
=> => sha256:27ee9a8250487842a26b1ffa1215982ba9ae 49.34MB / 49.34MB 14.2s
=> => extracting sha256:27ee9a8250487842a26b1ffa1215982ba9ae27010bce 3.0s
=> => extracting sha256:00f78834a2fef03250be89a7741ff39d0e6bbd860298 1.1s
=> => extracting sha256:cbc19164244e861d91ebd80a17e2c78c5be43a8059a3 3.3s
=> => extracting sha256:44fed46b68cf907ddf6afb6bc4211842f016a0876b0 10.3s
=> => extracting sha256:1599fa592074badc651117d15e0d7cc070b21d3dd9b6 0.4s
=> => extracting sha256:c5b14f510815b7cd08ec1fcfa5a5222e352011816f10 1.2s
=> => extracting sha256:7af818fe6364cc23e3187871665225f2ed1ffd6f965d 0.0s
=> [internal] load build context                                  0.1s
=> => transferring context: 284B                                   0.0s
=> [2/5] WORKDIR /app                                           0.4s
=> [3/5] COPY requirements.txt .                                0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt     9.8s
=> [5/5] COPY main.py .                                         0.1s
=> exporting to image                                           2.4s
=> => exporting layers                                           1.5s

```

8. List Docker Images: docker images

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	Info →	U	In Use	EXTRA
docker-demo:1.0	3b67be2e461d	224MB	55.1MB				U
hello-world:latest	5dd0d3e6e255	25.9kB	9.49kB				
my-docker-app:latest	cb904a724ddb	1.62GB	418MB				
nginx:latest	05b8cb60c354	253MB	69.2MB			U	
python:3.12-slim	09f7da3bc104	179MB	45.4MB				
redis:latest	298e5b3bc566	212MB	57.5MB				
srishti2902/my-nginx:1.0	05b8cb60c354	253MB	69.2MB			U	
ubuntu:latest	2260313b31c8	160MB	45.3MB			U	

9. Run Docker Container: docker run -d -p 5000:5000 --name my-container my-docker-app

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker run -d -p 5000:5000 --name my-container my-docker-app
8f3b05f459fb0bb87581cd4fb70f52da9c71a5ef5c2321b8b9c50d080bfd0459

```

10. Check Running Containers: docker ps

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
8f3b05f459fb	my-docker-app	"python main.py"	6 seconds ago	Up	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	my-container
7b82b6e69525	nginx	"/docker-entrypoint..."	14 minutes ago	Up	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	my-nginx

11. Check Docker Image Layers: docker history my-docker-app

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker history my-docker-app
```

IMAGE	SIZE	COMMENT	CREATED	CREATED BY
ae47b29b621b	0B	About a minute ago	CMD ["python" "main.py"]	buildkit.dockerfile.v0
<missing>	0B	About a minute ago	EXPOSE [5000/tcp]	buildkit.dockerfile.v0
<missing>	12.3kB	About a minute ago	COPY main.py . # buildkit	buildkit.dockerfile.v0
<missing>	15.6MB	About a minute ago	RUN /bin/sh -c pip install --no-cache-dir -r...	buildkit.dockerfile.v0
<missing>	12.3kB	About a minute ago	COPY requirements.txt . # buildkit	buildkit.dockerfile.v0
<missing>	8.19kB	About a minute ago	WORKDIR /app	buildkit.dockerfile.v0
<missing>	0B	2 weeks ago	CMD ["python3"]	buildkit.dockerfile.v0
<missing>	16.4kB	2 weeks ago	RUN /bin/sh -c set -eux; for src in idl	buildkit.dockerfile.v0
<missing>	75.9MB	2 weeks ago	RUN /bin/sh -c set -eux; wget -O pytho	buildkit.dockerfile.v0
<missing>	0B	2 weeks ago	ENV PYTHON_SHA256=1e66a7945a48390ee4c2a4	buildkit.dockerfile.v0
<missing>	0B	2 weeks ago	ENV PYTHON_VERSION=3.13.15	buildkit.dockerfile.v0
<missing>	0B	2 weeks ago	ENV GPG_KEY=7169605F62C751356D054A26A821	buildkit.dockerfile.v0
<missing>	19.9MB	2 weeks ago	RUN /bin/sh -c set -eux; apt-get update	buildkit.dockerfile.v0
<missing>	0B	2 weeks ago	ENV PATH=/usr/local/bin:/usr/local/sbin:	buildkit.dockerfile.v0
<missing>	694MB	2 weeks ago	RUN /bin/sh -c set -ex; apt-get update;	buildkit.dockerfile.v0
<missing>	202MB	2 weeks ago	RUN /bin/sh -c set -eux; apt-get update	buildkit.dockerfile.v0
<missing>	65MB	2 weeks ago	RUN /bin/sh -c set -eux; apt-get update	buildkit.dockerfile.v0
<missing>		2 weeks ago	# debian.sh --arch 'amd64' out/ 'trixie'	

12. Stop Docker Container: docker stop my-container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
8f3b05f459fb	my-docker-app	"python main.py"	About a minute ago	Up	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	my-container
7b82b6e69525	nginx	"/docker-entrypoint..."	15 minutes ago	Up	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	my-nginx

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker stop my-container
```

```
my-container
```

13. Check All Containers: docker ps -a

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	N
8f3b05f459fb	my-docker-app	"python main.py"	About a minute ago	Exited (137) 14 seconds ago		m
7b82b6e69525	nginx	"/docker-entrypoint..."	16 minutes ago	Up 16 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	m
39cc1e7f5e4f	ubuntu	"/bin/bash"	53 minutes ago	Exited (1) 53 minutes ago		s
c61b7cc14518	nginx	"/docker-entrypoint..."	23 hours ago	Exited (255) 58 minutes ago	80/tcp	v
45a04f4d0e37	docker-demo:1.0	"python app.py"	2 days ago	Exited (255) 24 hours ago	0.0.0.0:5000->5000/tcp	d

```
docker-demo-container
```

14. Start Docker Container: docker start my-container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker start my-container
my-container
```

15. View Container Logs: docker logs my-container

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker logs my-container
```

```
* Serving Flask app 'main'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.3:5000
Press CTRL+C to quit
172.17.0.1 - - [09/Sep/2026 13:03:52] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [09/Sep/2026 13:03:52] "GET /favicon.ico HTTP/1.1" 404 -
* Serving Flask app 'main'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.3:5000
Press CTRL+C to quit
```

Git Commands

18. Initialize Git Repository: git init

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warni
ng,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName fal
se"

```

19. Check Git Status: git status

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    commands.txt
    main.py
    observations.txt
    requirements.txt

nothing added to commit but untracked files present (use "git add" to track)

```

20. Add Project Files: git add .

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Dockerfile
    commands.txt
    main.py
    observations.txt
    requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .

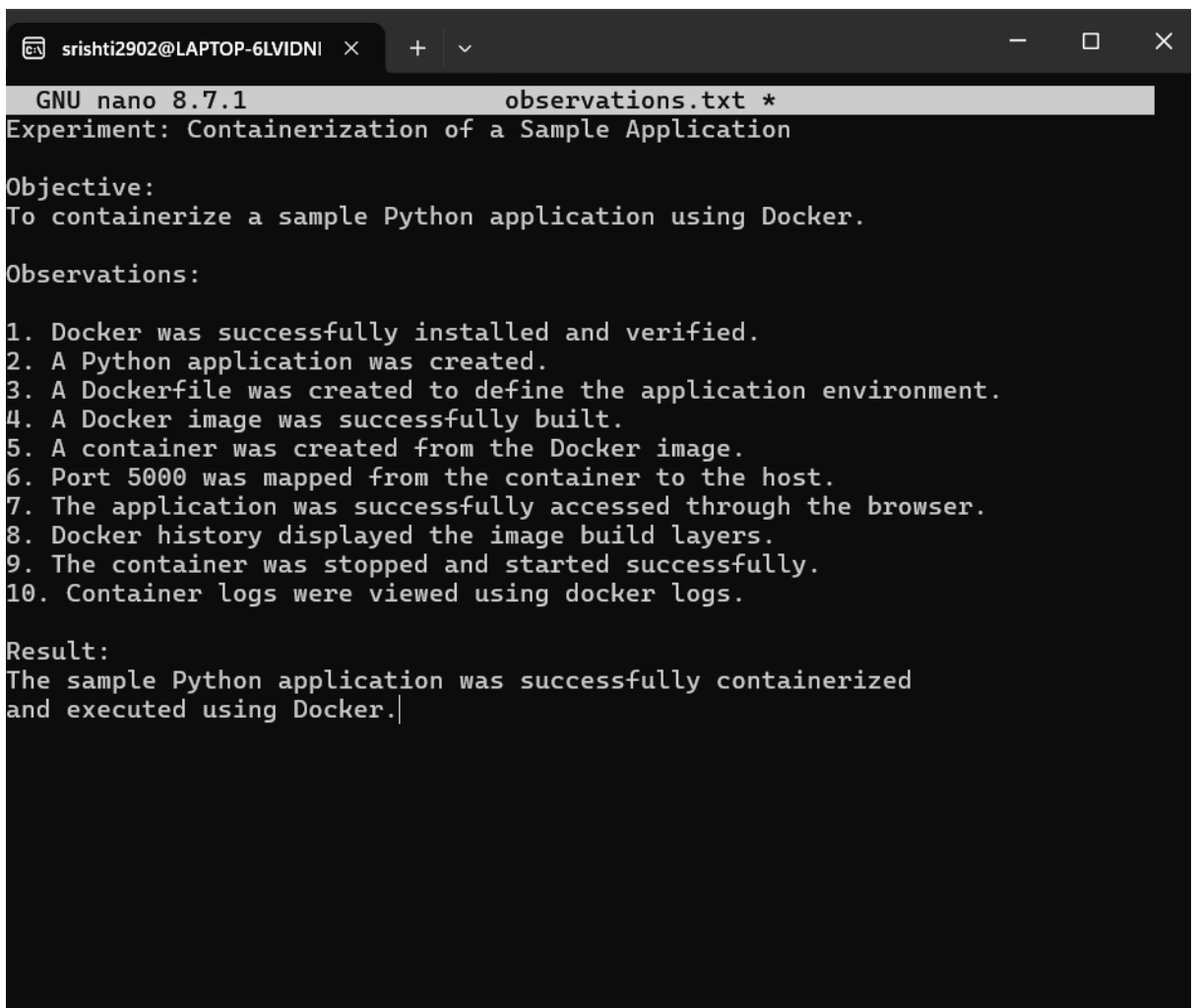
```

21. Create Initial Commit: git commit -m "Initial Docker project"

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Initial Docker project"
[master (root-commit) 9ec3c01] Initial Docker project
5 files changed, 67 insertions(+)
create mode 100644 Dockerfile
create mode 100644 commands.txt
create mode 100644 main.py
create mode 100644 observations.txt
create mode 100644 requirements.txt
```

22. Update Observations File: nano observations.txt

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano observations.txt
```



```
GNU nano 8.7.1 observations.txt *
Experiment: Containerization of a Sample Application

Objective:
To containerize a sample Python application using Docker.

Observations:

1. Docker was successfully installed and verified.
2. A Python application was created.
3. A Dockerfile was created to define the application environment.
4. A Docker image was successfully built.
5. A container was created from the Docker image.
6. Port 5000 was mapped from the container to the host.
7. The application was successfully accessed through the browser.
8. Docker history displayed the image build layers.
9. The container was stopped and started successfully.
10. Container logs were viewed using docker logs.

Result:
The sample Python application was successfully containerized
and executed using Docker.
```

23. Add Updated Files: git add .

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano observations.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
git commit -m "Updated Docker experiment observations"
On branch master
nothing to commit, working tree clean
```

24. Create Second Commit: git commit -m "Updated Docker experiment observations"

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Initial Docker
project"
[master (root-commit) 9ec3c01] Initial Docker project
5 files changed, 67 insertions(+)
create mode 100644 Dockerfile
create mode 100644 commands.txt
create mode 100644 main.py
create mode 100644 observations.txt
create mode 100644 requirements.txt

```

25. View Commit History: git log --oneline , git status

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline
9ec3c01 (HEAD -> master) Initial Docker project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch master
nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$

```

2. Prepare a CI/CD workflow

Prepare a CI/CD workflow report describing the pipeline architecture, stages, automation results, and observations with supporting screenshots.

1. **Check Git Installation:** git --version
2. **Configure Git Username:** git config --global user.name "Your Name"
3. **Configure Git Email:** git config --global user.email "your-email@example.com"
4. **Check Git Configuration:** git config --global --list
5. **Initialize Git Repository:** git init
6. **Check Git Status:** git status
7. **Add Project Files:** git add .
8. **Create Git Commit:** git commit -m "Initial Docker project"
9. **Create Observations File:** nano observations.txt
10. **View Commit History:** git log --oneline

11. **Add GitHub Remote:** `git remote add origin YOUR_GITHUB_REPOSITORY_URL`
12. **Check GitHub Remote:** `git remote -v`
13. **Rename Branch to Main:** `git branch -M main`
14. **Pull Remote Changes:** `git pull origin main --rebase`
15. **Push Project to GitHub:** `git push -u origin main`
16. **Create GitHub Actions Directory:** `mkdir -p .github/workflows`
17. **Create CI/CD Workflow File:** `nano .github/workflows/docker-ci.yml`
18. **Check Workflow File:** `ls .github/workflows`
19. **Display Workflow Configuration:** `cat .github/workflows/docker-ci.yml`
20. **Add CI/CD Workflow:** `git add .`
21. **Commit CI/CD Workflow:** `git commit -m "Added Docker CI/CD workflow"`
22. **Push CI/CD Workflow:** `git push`
23. **Test Application:** `curl http://localhost:5000`
24. **View Git Commit History:** `git log --oneline`

```
srishti2902@LAPTOP-6LVIDNI x + v - □ ×
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cd ~/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile  commands.txt  main.py  observations.txt  requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch master
nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker-cicd-project
docker-cicd-project: command not found
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cd ~/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote add origin ht
tps://github.com/srishti2902/Devops-lab
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote -v
origin  https://github.com/srishti2902/Devops-lab (fetch)
origin  https://github.com/srishti2902/Devops-lab (push)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch -M main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Srishti
Password for 'https://Srishti@github.com':
remote: Invalid username or token. Password authentication is not supp
orted for Git operations.
fatal: Authentication failed for 'https://github.com/srishti2902/Devop
s-lab/'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Srishti
Password for 'https://Srishti@github.com':
remote: Invalid username or token. Password authentication is not supp
orted for Git operations.
fatal: Authentication failed for 'https://github.com/srishti2902/Devop
s-lab/'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Srishti2902
Password for 'https://Srishti2902@github.com':
remote: Invalid username or token. Password authentication is not supp
orted for Git operations.
fatal: Authentication failed for 'https://github.com/srishti2902/Devop
s-lab/'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Srishti2902
Password for 'https://Srishti2902@github.com':
remote: Invalid username or token. Password authentication is not supp
orted for Git operations.
fatal: Authentication failed for 'https://github.com/srishti2902/Devop
```

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git pull origin main --rebase
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (10/10), 4.38 MiB | 3.01 MiB/s, done.
From https://github.com/srishti2902/Devops-lab
* branch          main          -> FETCH_HEAD
* [new branch]     main          -> origin/main
Successfully rebased and updated refs/heads/main.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 1.31 KiB | 1.31 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/srishti2902/Devops-lab
   1aca16..db2aaf6  main -> main
branch 'main' set up to track 'origin/main'.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ mkdir -p .github/workflows
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls -la
total 4868
drwxr-xr-x  4 srishti2902 srishti2902  4096 Sep  9 13:49 .
drwxr-xr-x 13 srishti2902 srishti2902  4096 Sep  9 13:12 ..
drwxr-xr-x  7 srishti2902 srishti2902  4096 Sep  9 13:49 .git
drwxr-xr-x  3 srishti2902 srishti2902  4096 Sep  9 13:49 .github
-rw-r--r--  1 srishti2902 srishti2902   164 Sep  9 13:48 Dockerfile
-rw-r--r--  1 srishti2902 srishti2902 3669414 Sep  9 13:48 'LAB2 TASK2.pdf'
-rw-r--r--  1 srishti2902 srishti2902  743149 Sep  9 13:48 'Lab2 (1).docx'
-rw-r--r--  1 srishti2902 srishti2902    12 Sep  9 13:48 README.md
-rw-r--r--  1 srishti2902 srishti2902   301 Sep  9 13:48 commands.txt
-rw-r--r--  1 srishti2902 srishti2902 526460 Sep  9 13:48 lab2.docx
-rw-r--r--  1 srishti2902 srishti2902   197 Sep  9 13:48 main.py
-rw-r--r--  1 srishti2902 srishti2902   762 Sep  9 13:48 observatio
```

```
srishti2902@LAPTOP-6LVIDNI x + v
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls .github/workflows
docker-ci.yml
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cat .github/workflows/docker-ci.yml
name: Docker CI/CD

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Build Docker image
        run: docker build -t my-docker-app .

      - name: Run Docker container
        run: |
          docker run -d -p 5000:5000 --name my-container my-docker-app
          sleep 5

      - name: Test application
        run: curl http://localhost:5000

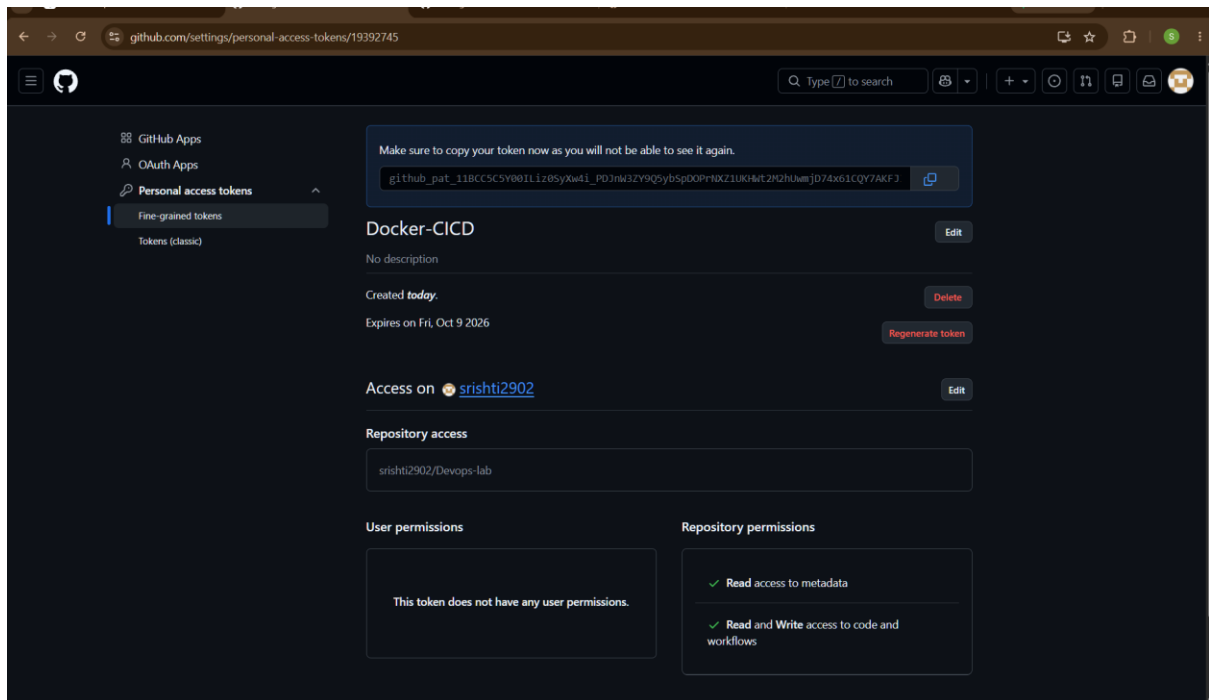
      - name: Stop container
        run: docker stop my-container
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

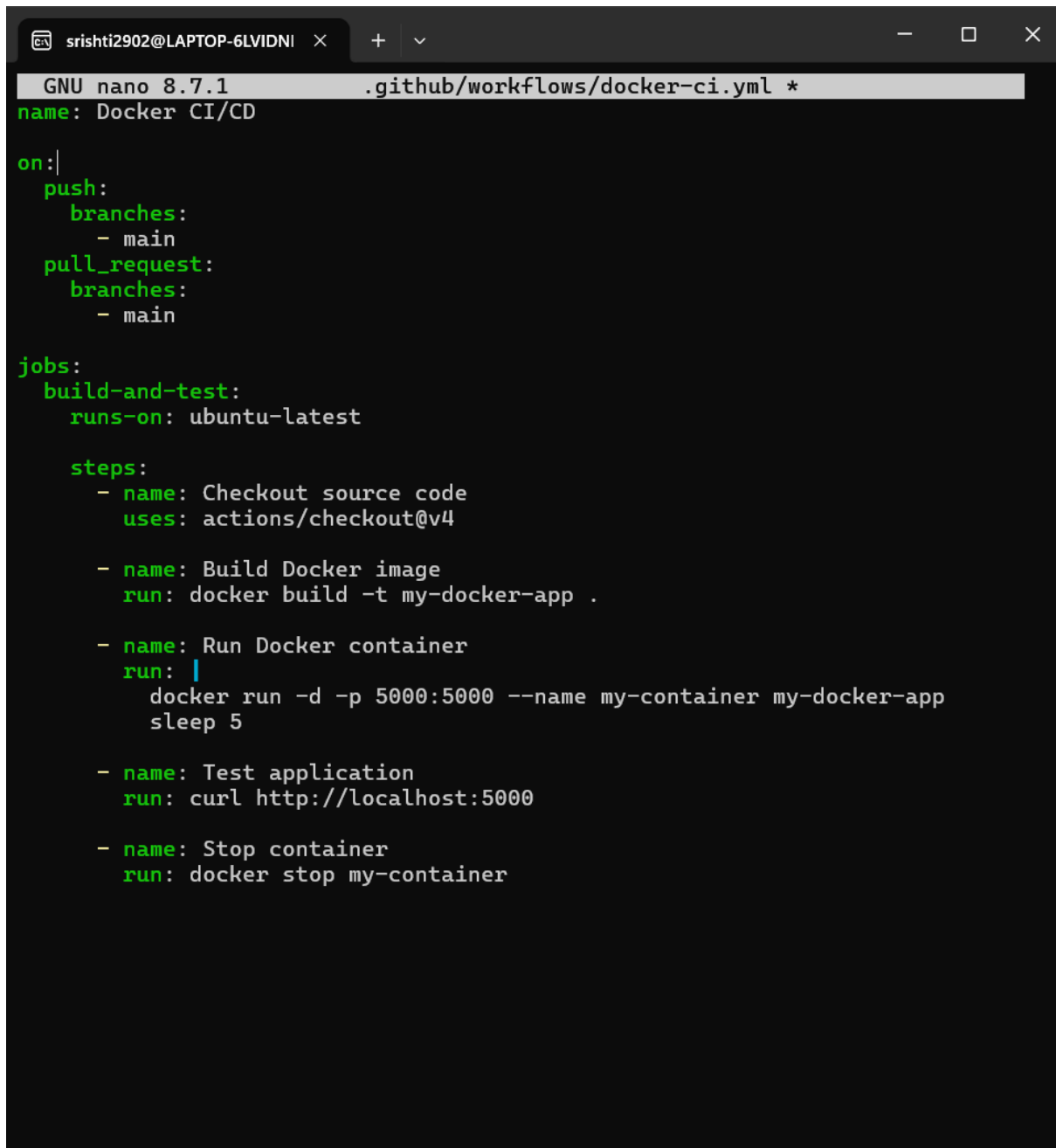
Untracked files:
  (use "git add <file>..." to include in what will be committed)
.github/
```

```
srishti2902@LAPTOP-6LVIDNI x + v - □ ×

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  .github/

nothing added to commit but untracked files present (use "git add" to
track)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Added Doc
ker CI/CD workflow"
[main 9e7f00c] Added Docker CI/CD workflow
 1 file changed, 31 insertions(+)
 create mode 100644 .github/workflows/docker-ci.yml
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 647 bytes | 647.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/srishti2902/Devops-lab
 ! [remote rejected] main -> main (refusing to allow a Personal Access
Token to create or update workflow '.github/workflows/docker-ci.yml'
without 'workflow' scope)
error: failed to push some refs to 'https://github.com/srishti2902/Dev
ops-lab'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 647 bytes | 129.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/srishti2902/Devops-lab
 db2aaf6..9e7f00c main -> main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$
```





```
GNU nano 8.7.1 .github/workflows/docker-ci.yml *
name: Docker CI/CD

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Build Docker image
        run: docker build -t my-docker-app .

      - name: Run Docker container
        run: |
          docker run -d -p 5000:5000 --name my-container my-docker-app
          sleep 5

      - name: Test application
        run: curl http://localhost:5000

      - name: Stop container
        run: docker stop my-container
```

3. Automate application deployment

Automate application deployment using Jenkins by configuring build triggers, post-build actions, and deployment steps.

1. **Check Docker Installation:** `docker --version`
2. **Check Jenkins Installation:** `jenkins --version`
3. **Check Jenkins Status:** `sudo systemctl status jenkins`
4. **Start Jenkins:** `sudo systemctl start jenkins`
5. **Enable Jenkins at Startup:** `sudo systemctl enable jenkins`
6. **Check Docker Containers:** `docker ps`
7. **Build Docker Image:** `docker build -t my-docker-app .`
8. **Stop Previous Container:** `docker stop my-container || true`
9. **Remove Previous Container:** `docker rm my-container || true`
10. **Deploy New Docker Container:** `docker run -d -p 5000:5000 --name my-container my-docker-app`
11. **Verify Running Container:** `docker ps`
12. **View Container Logs:** `docker logs my-container`
13. **Test Deployed Application:** `curl http://localhost:5000`
14. **Check Git Installation:** `git --version`
15. **Check Git Repository Status:** `git status`
16. **View Git Commit History:** `git log --oneline`

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue



+ New Item

Build History

Build Queue



No builds in the queue.

Build Executor Status

0/2

Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job



Set up a distributed build

Set up an agent

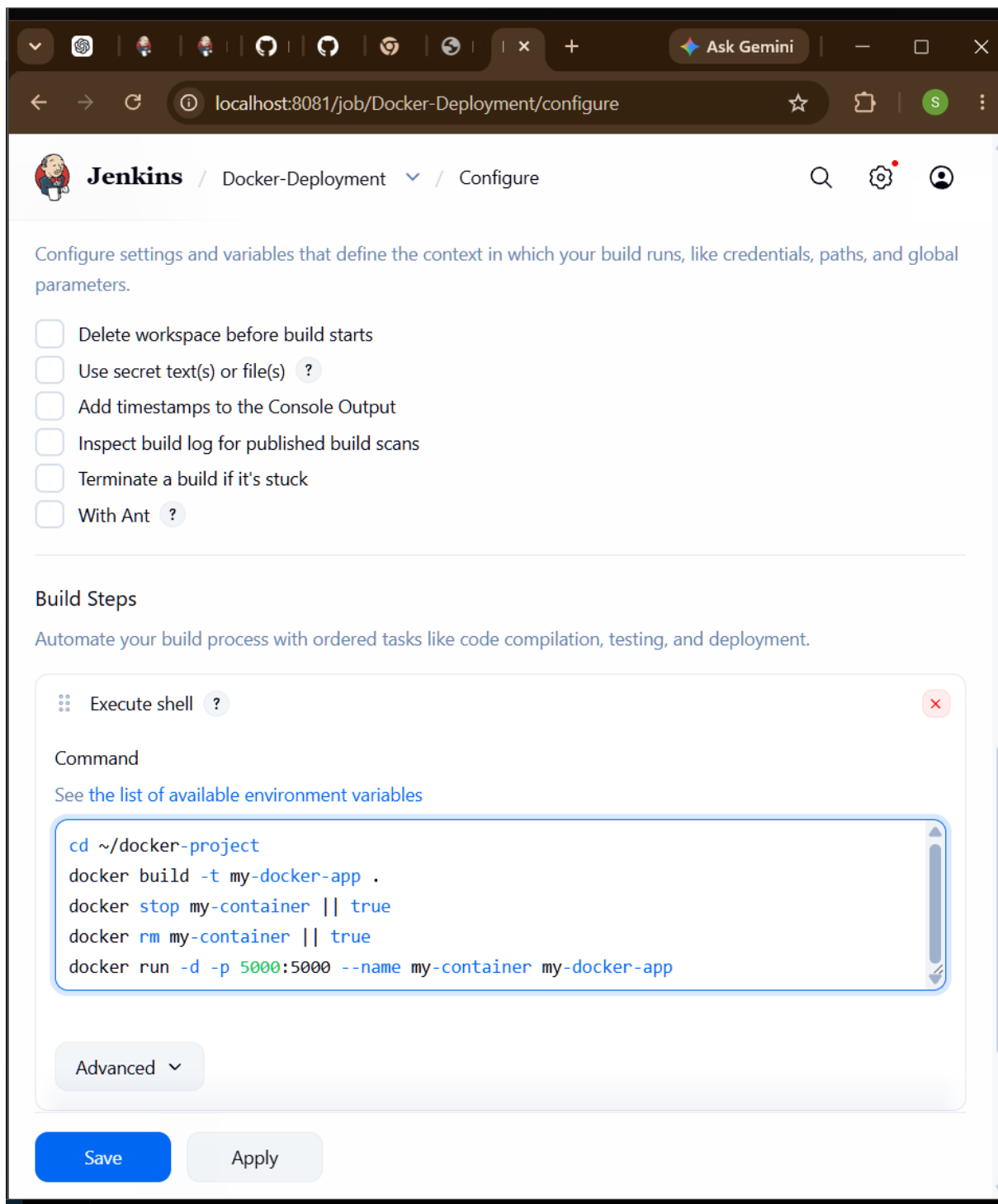


Configure a cloud



Learn more about distributed builds





← → ↺ ↻

localhost:8081/job/Docker-Deployment/

☆

⌵

S

⋮

 **Jenkins** / Docker-Deployment



 Status

</> Changes

 Workspace

 Build Now

 Delete Project

 Configure

 Rename

Builds >

...

No builds

Docker-Deployment

 Add description

Permalinks

REST API

Jenkins 2.568.3

```
srishti2902@LAPTOP-6LVIDNI x + v - □ ×

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker --version
Docker version 29.7.2, build a7dcaa6
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker rm -f jenkins 2>/dev/null || true
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker volume create jenkins_home
jenkins_home
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker run -d --name jenkins -p 8081:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
d097c8512a96: Pull complete
13045988ecda: Pull complete
91db25d7b900: Pull complete
549414223487: Pull complete
009bab87a9ea: Pull complete
c739d2dd7782: Pull complete
f06c632bcf59: Pull complete
e8d1cd9c12ca: Pull complete
7d8676c4f825: Pull complete
88d27572059a: Pull complete
ba74585817cb: Pull complete
44136fa355b3: Already exists
272717b30638: Download complete
Digest: sha256:c1e4c349365f6d16d88595b2c5f7e8ff39b8ae1d061f62420bac193b4b9616d0
Status: Downloaded newer image for jenkins/jenkins:lts
14a5049451a29759105ee091d7552b50045b9e52d28a92467868fb7a02bd5bfb
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED
STATUS        PORTS          NAMES
14a5049451a2   jenkins/jenkins:lts   "/usr/bin/tini -- /u..."   5 second
s ago        Up 4 seconds   0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   jenkins
8f3b05f459fb   ae47b29b621b         "python main.py"           2 hours
ago         Up 2 hours     0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   my-container
7b82b6e69525   nginx          "/docker-entrypoint..."   2 hours
ago         Up 2 hours     0.0.0.0:8080->80/tcp, [::]:8080->80/tcp       my-nginx
```

```
srishti2902@LAPTOP-6LVIDNI x + v
ago Up 2 hours 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
my-container
7b82b6e69525 nginx "/docker-entrypoint..." 2 hours
ago Up 2 hours 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
my-nginx
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker exec jenkins cat
/var/jenkins_home/secrets/initialAdminPassword
66aec1a9f4c44a7293c936dfd73c4da7
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker ps
CONTAINER ID IMAGE PORTS COMMAND CREATED
STATUS NAMES
14a5049451a2 jenkins/jenkins:lts "/usr/bin/tini -- /u..." 17 minut
es ago Up 17 minutes 0.0.0.0:50000->50000/tcp, [::]:50000->50000/t
cp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp jenkins
8f3b05f459fb ae47b29b621b "python main.py" 2 hours
ago Up 2 hours 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
my-container
7b82b6e69525 nginx "/docker-entrypoint..." 2 hours
ago Up 2 hours 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
my-nginx
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ curl http://localhost:50
00
Hello! My Docker application is running.srishti2902@LAPTOP-6LVIDNHT:~/
docker-project$ docker ps
CONTAINER ID IMAGE PORTS COMMAND CREATED
STATUS NAMES
14a5049451a2 jenkins/jenkins:lts "/usr/bin/tini -- /u..." 17 minut
es ago Up 17 minutes 0.0.0.0:50000->50000/tcp, [::]:50000->50000/t
cp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp jenkins
8f3b05f459fb ae47b29b621b "python main.py" 2 hours
ago Up 2 hours 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
my-container
7b82b6e69525 nginx "/docker-entrypoint..." 2 hours
ago Up 2 hours 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
my-nginx
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ curl http://localhost:50
00
Hello! My Docker application is running.srishti2902@LAPTOP-6LVIDNHT:~/
docker-project$ docker --version
Docker version 29.7.2, build a7dcaa6
```

4. Install and configure Git and Jenkins

Install and configure Git on Ubuntu Linux, verify the installation, and set up global Git configuration (user name and email).

1. **Update Ubuntu Packages:** `sudo apt update`

2. **Install Git:** `sudo apt install git -y`

3. **Verify Git Installation:** `git --version`

4. **Configure Git Username:** `git config --global user.name "Srishti Mishra"`
5. **Configure Git Email:** `git config --global user.email "your-email@example.com"`
6. **Verify Git Configuration:** `git config --global --list`
7. **Check Git Username:** `git config --global user.name`
8. **Check Git Email:** `git config --global user.email`
9. **Check Git Status:** `git status`

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:6 https://pkg.jenkins.io/debian-stable binary/ Release
154 packages can be upgraded. Run 'apt list --upgradable' to see them.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ sudo apt install git -y
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 154
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git --version
git version 2.53.0
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git config --global user
.name "Srishti Mishra"
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git config --global user
.email "mishrasrishti309@gmail.com"
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git config --global --li
st
user.name=Srishti Mishra
user.email=mishrasrishti309@gmail.com
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git config --global user
.name
Srishti Mishra
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git config --global user
.email
mishrasrishti309@gmail.com
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ |
```

5. Set up version control using Git

initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

1. Navigate to Project Directory: `cd ~/docker-project`

2. List Project Files: `ls`

3. Initialize Git Repository: `git init`

4. Check Repository Status: `git status`

5. Stage All Changes: `git add .`

6. Commit Changes: `git commit -m "Initial project commit"`

7. View Commit History: `git log --oneline`

8. Create Development Branch: `git branch development`

9. List Branches: `git branch`

10. Switch to Development Branch: `git switch development`

11. Stage Updated Files: `git add .`

12. Commit Branch Changes: `git commit -m "Updated application"`

13. Switch to Main Branch: `git switch main`

14. Merge Development Branch: `git merge development`

15. View Branch and Commit History: `git log --oneline --graph --all`

16. Add GitHub Remote: `git remote add origin
YOUR_GITHUB_REPOSITORY_URL`

17. Verify Remote Repository: `git remote -v`

18. Rename Branch to Main: `git branch -M main`

19. Push Repository to GitHub: `git push -u origin main`

20. Verify Repository Status: `git status`

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cd ~/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile      README.md      main.py
'LAB2 TASK2.pdf'  commands.txt  observations.txt
'Lab2 (1).docx'  lab2.docx     requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git init
Reinitialized existing Git repository in /home/srishti2902/docker-project/.git/
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Initial project commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline
9e7f00c (HEAD -> main, origin/main) Added Docker CI/CD workflow
db2aaf6 Initial Docker project
1acaa16 Add files via upload
7e99c4c Add files via upload
f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch development
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
  development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git switch development
Switched to branch 'development'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
* development
  main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
```

```
srishti2902@LAPTOP-6LVIDNI x + v
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Initial p
project commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline
9e7f00c (HEAD -> main, origin/main) Added Docker CI/CD workflow
db2aaf6 Initial Docker project
1acaa16 Add files via upload
7e99c4c Add files via upload
f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch development
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
  development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git switch development
Switched to branch 'development'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
* development
  main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano main.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch development
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory
)
       modified:   main.py

no changes added to commit (use "git add" and/or "git commit -a")
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Updated a
pplication"
[development 5a81cb8] Updated application
 1 file changed, 1 insertion(+), 2 deletions(-)
```

```
srishti2902@LAPTOP-6LVIDNI x + v
Fast-forward
main.py | 3 +--
1 file changed, 1 insertion(+), 2 deletions(-)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline --graph --all
* 5a81cb8 (HEAD -> main, development) Updated application
* 9e7f00c (origin/main) Added Docker CI/CD workflow
* db2aaf6 Initial Docker project
* 1acaa16 Add files via upload
* 7e99c4c Add files via upload
* f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote add origin https://github.com/srishti2902/Devops-lab
error: remote origin already exists.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote -v
origin https://github.com/srishti2902/Devops-lab (fetch)
origin https://github.com/srishti2902/Devops-lab (push)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch -M main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 298 bytes | 149.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/srishti2902/Devops-lab
  9e7f00c..5a81cb8 main -> main
branch 'main' set up to track 'origin/main'.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline --graph --all
* 5a81cb8 (HEAD -> main, origin/main, development) Updated application
```

```

branch 'main' set up to track 'origin/main'.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
  development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline --graph --all
* 5a81cb8 (HEAD -> main, origin/main, development) Updated application
* 9e7f00c Added Docker CI/CD workflow
* db2aaf6 Initial Docker project
* 1acaa16 Add files via upload
* 7e99c4c Add files via upload
* f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote -v
origin  https://github.com/srishti2902/Devops-lab (fetch)
origin  https://github.com/srishti2902/Devops-lab (push)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$

```

5.Set up version control using Git

initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

```
srishti2902@LAPTOP-6LVIDNI x + v - □ ×
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cd ~/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
  Dockerfile      README.md      main.py
'LAB2 TASK2.pdf'  commands.txt   observations.txt
'Lab2 (1).docx'   lab2.docx      requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git init
Reinitialized existing Git repository in /home/srishti2902/docker-project/.git/
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Initial project commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline
5a81cb8 (HEAD -> main, origin/main, development) Updated application
9e7f00c Added Docker CI/CD workflow
db2aaf6 Initial Docker project
1acaa16 Add files via upload
7e99c4c Add files via upload
f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch development
fatal: a branch named 'development' already exists
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
  development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git switch development
Switched to branch 'development'
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
* development
  main
```

```
srishti2902@LAPTOP-6LVIDNI x + v - □ ×
plication"
[development a9862d3] Update application
 1 file changed, 1 insertion(+), 1 deletion(-)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline
a9862d3 (HEAD -> development) Update application
5a81cb8 (origin/main, main) Updated application
9e7f00c Added Docker CI/CD workflow
db2aaf6 Initial Docker project
1acaa16 Add files via upload
7e99c4c Add files via upload
f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
  development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch
  development
* main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git log --oneline --graph
h --all
* a9862d3 (development) Update application
* 5a81cb8 (HEAD -> main, origin/main) Updated application
* 9e7f00c Added Docker CI/CD workflow
* db2aaf6 Initial Docker project
* 1acaa16 Add files via upload
* 7e99c4c Add files via upload
* f93a1f5 Initial commit
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote add origin ht
tps://github.com/srishti2902/Devops-lab
error: remote origin already exists.
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git remote -v
origin https://github.com/srishti2902/Devops-lab (fetch)
origin https://github.com/srishti2902/Devops-lab (push)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git branch -M main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push -u origin main
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
branch 'main' set up to track 'origin/main'.
Everything up-to-date
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ |
```

6. Implement a basic Jenkins CI/CD pipeline

Implement a basic Jenkins CI/CD pipeline that pulls source code from the GitHub repository and triggers an automated build.



Jenkins



+ New Item

📁 Build History

Build Queue



No builds in the queue.

Build Executor Status

0/2

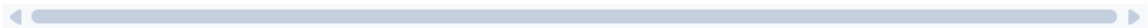


📝 Add description

All

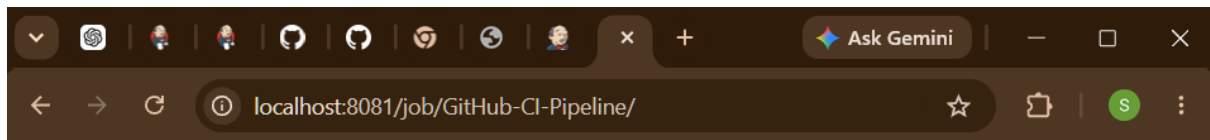


S	W	Name ↓	Last Success	Last Failure	Last Duration	
❌	☁️	Docker-Deployment	N/A	17 sec #9	0.32 sec	▶️



Icon: S M L





-  Status
-  Changes
-  Workspace
-  Build Now
-  Delete Project
-  Git Polling Log
-  Configure
-  Rename

Builds > ⋮

Pending

 #1 

Finished waiting

GitHub-CI-Pipeline

 Add description

Permalinks



Jenkins

/ GitHub-CI-Pipeline



/ #1



Status

</> Changes



Console Output



Delete build '#1'



Polling Log



Git Build Data



Edit build information



#1 (Sep 9, 2026, 3:51:08 PM)



Add description

Keep this build forever



Started by an SCM change

Started 1 min 21 sec ago

Took 10 sec



Revision: 5a81cb80d3bd5ade1398b7a20bdd0e6b516745ae

Repository: <https://github.com/srishti2902/Devops-lab>

- refs/remotes/origin/main



No changes.

REST API

Jenkins 2.568.3

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cd ~/docker-project
nano main.py
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ^C
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add .
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Test Jenkins CI trigger"
[main 7ff62ae] Test Jenkins CI trigger
 1 file changed, 1 insertion(+), 1 deletion(-)
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Test Jenkins CI trigger"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ |
```

7. Create a Jenkins file defining pipeline stages

Create a Jenkinsfile defining pipeline stages (build, test, deploy) and demonstrate a declarative or scripted pipeline execution.

```
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cd ~/docker-project
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ^C
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile      README.md      main.py
'LAB2 TASK2.pdf'  commands.txt  observations.txt
'Lab2 (1).docx'   lab2.docx     requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ ls
Dockerfile      README.md      main.py
'LAB2 TASK2.pdf'  commands.txt  observations.txt
'Lab2 (1).docx'   lab2.docx     requirements.txt
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ nano Jenkinsfile

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ cat Jenkinsfile
pipeline {
    agent any

    stages {
        stage('Build') {
            steps {
                echo 'Building the application...'
                sh 'docker build -t my-docker-app .'
            }
        }

        stage('Test') {
            steps {
                echo 'Testing the application...'
                sh 'docker image inspect my-docker-app'
            }
        }

        stage('Deploy') {
            steps {
                echo 'Deploying the application...'
                sh 'docker stop my-container || true'
                sh 'docker rm my-container || true'
                sh 'docker run -d -p 5000:5000 --name my-container my-
docker-app'
            }
        }
    }
}
```



Jenkins

/ Docker-Pipeline

/ Configure



Configure



General



Source Code Management



Triggers



Environment



Build Steps



Post-build Actions

General

Enabled



Description

Plain text [Preview](#)



Discard old builds ?



GitHub project

Save

Apply



Advanced ▼

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/srishti2902/Devops-lab

Credentials ?

- none -



+ Add

Advanced ▼

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

Save

Apply



Status



Changes



Workspace



Build Now



Delete Project



Git Polling Log



Configure



Rename

Builds >



No builds

Docker-Pipeline



Add description

Permalinks

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git add Jenkinsfile
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git commit -m "Add Jenkins declarative pipeline"
[main 8e0a564] Add Jenkins declarative pipeline
 1 file changed, 38 insertions(+)
 create mode 100644 Jenkinsfile
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ git push
Username for 'https://github.com': Docker-CICD
Password for 'https://Docker-CICD@github.com':
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 791 bytes | 263.00 KiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To https://github.com/srishti2902/Devops-lab
   5a81cb8..8e0a564  main -> main
srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
14a5049451a2	jenkins/jenkins:lts	"/usr/bin/tini -- /u..."	About an hour ago	Up About an hour	0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp	jenkins
8f3b05f459fb	ae47b29b621b	"python main.py"	3 hours ago	Up 3 hours	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	my-container
7b82b6e69525	nginx	"/docker-entrypoint..."	3 hours ago	Up 3 hours	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	my-nginx

```

srishti2902@LAPTOP-6LVIDNHT:~/docker-project$ curl http://localhost:5000
Hello! My Docker application is running.srishti2902@LAPTOP-6LVIDNHT:~/
docker-project$ stage('Build')

```